Problem 2

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Problem 1 *Find the limit and justify your answer.*

(a) $\lim_{x\to 0} |x|$

Explanation. Remember $|x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$

We first compute the one-sided limit: $\lim_{x\to 0^+} |x| = \lim_{x\to 0^+} x = 0$.

Next, we compute the other one-sided limit:

$\lim_{x\to 0^-} |x| = \lim_{x\to 0^-} (-x) = \left(\lim_{x\to 0^-} (-1)\right)\left(\lim_{x\to 0^-} x\right) = (-1)(0) = 0.$

Therefore $\lim_{x\to 0^-} |x| = 0 = \lim_{x\to 0^+} |x|$. Hence, $\lim_{x\to 0} |x| = 0$.

(b) $\lim_{x\to 2} \ln\left(\sin(x-2) + e^x \sin\left(\frac{\pi x}{4}\right)\right)$

Explanation.

$$\begin{aligned}
\lim_{x\to 2} \ln\left(\sin(x-2) + e^x \sin\left(\frac{\pi x}{4}\right)\right) &= \ln\left(\lim_{x\to 2}\left(\sin(x-2) + e^x \sin\left(\frac{\pi x}{4}\right)\right)\right) \\
&= \ln\left(\lim_{x\to 2} \sin(x-2) + \lim_{x\to 2}\left(e^x \sin\left(\frac{\pi x}{4}\right)\right)\right) \\
&= \ln\left(\sin\left(\lim_{x\to 2}(x-2)\right) + \left(\lim_{x\to 2} e^x\right)\left(\lim_{x\to 2} \sin\left(\frac{\pi x}{4}\right)\right)\right) \\
&= \ln\left(\sin(0) + \left(\lim_{x\to 2} e^x\right) \sin\left(\lim_{x\to 2} \frac{\pi x}{4}\right)\right) \\
&= \ln\left(0 + e^2 \sin\left(\frac{2\pi}{4}\right)\right) \\
&= \ln\left(e^2 \sin\left(\frac{\pi}{2}\right)\right) = \ln\left(e^2(1)\right) \\
&= \ln\left(e^2\right) = 2
\end{aligned}$$

(c) $\lim_{x\to 1} \dfrac{3 + 2\cos\left(\frac{\pi x}{3}\right)}{4x - 2x^3}$

Explanation.

$$
\begin{aligned}
\lim_{x\to 1} \frac{3+2\cos\left(\frac{\pi x}{3}\right)}{4x-2x^3} &= \frac{\lim\limits_{x\to 1}\left(3+2\cos\left(\dfrac{\pi x}{3}\right)\right)}{\lim\limits_{x\to 1}\left(4x-2x^3\right)} \\
&= \frac{\lim\limits_{x\to 1}(3)+\lim\limits_{x\to 1}\left(2\cos\left(\dfrac{\pi x}{3}\right)\right)}{4(1)-2(1)^3} \\
&= \frac{3+\left(\lim\limits_{x\to 1} 2\right)\left(\lim\limits_{x\to 1}\cos\left(\dfrac{\pi x}{3}\right)\right)}{4-2(1)} \\
&= \frac{3+2\cos\left(\lim\limits_{x\to 1}\dfrac{\pi x}{3}\right)}{4-2} \\
&= \frac{3+2\cos\left(\frac{\pi}{3}\right)}{2} = \frac{3+2\left(\frac{1}{2}\right)}{2} \\
&= \frac{3+1}{2} = 2.
\end{aligned}
$$